

Geyser Formation by Release of Entrapped Air from Horizontal Pipe into Vertical Shaft

Jing Cong¹; S. N. Chan²; and Joseph H. W. Lee, F.ASCE³

Abstract: Geysers are explosive eruptions of air-water mixture from manholes in drainage systems. When the design capacity of storm water drainage systems is exceeded during heavy rainfall, rapid inflows can lead to air-water interactions that give rise to geysers, causing damages and threatening human lives. Over the past two decades, the dynamics of air pockets in vertical shafts and horizontal pipes has been extensively studied. Although the role of entrapped air in causing large pressure transients is revealed, the mechanism of geyser formation remains elusive primarily because of the lack of detailed observations. A comprehensive series of experiments has been performed on a physical model of a simplified drainage system, which consists of a vertical riser and a horizontal pipe with diameter D connected to a constant head tank. The system is filled with water; an air pocket is then released into the horizontal pipe and the trajectories of the air pockets in horizontal pipe and vertical riser are measured by videos and a high speed camera. Pressures are measured using pressure transducers near the pipe end and at the bottom of the riser. Parameters considered include riser diameter (D_r), upstream head (H_0), and initial air pocket volume (V_{air}). Without an external pressure head, the air pocket migration in a water-filled vertical riser is found to be similar to a slug flow. The rising velocity of the air pocket relative to the free surface (v_{net}) is nearly constant and close to the speed of a Taylor bubble; no geyser is observed. When an external pressure head is applied to the horizontal pipe, two types of flow are observed. For large riser diameter and small air volume, the rise of the air pocket in the vertical riser resembles a Taylor bubble, with an air pressure equal to the hydrostatic pressure of the water column above. The air breaks within the riser and no geyser is observed. For small riser diameter and large air volume, the horizontal air pocket shoots past the riser junction and only partially enters the shaft; the vertical air pocket migration is very different from that of a Taylor bubble. The air pocket pressure is found to be significantly higher than the hydrostatic value, resulting in rapid acceleration of air and water, and jetting out of air-water mixture from the riser top. Experimental results show that geysers are more likely to occur for small risers ($D_r/D \leq 0.62$) and large air volumes $\{V_{\text{air}}/[(\pi D_r^2/4)H_0] \geq 3.42\}$. DOI: 10.1061/(ASCE)HY.1943-7900.0001332. © 2017 American Society of Civil Engineers.

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Introduction

Geysers are explosive eruptions of air-water mixture from manholes in drainage systems. Extreme rainfall intensity increases due to climate change (Lenderink et al. 2011), and with urbanization, the design capacity of storm water drainage systems in many cities is often exceeded (Konrad 2003). Rapid inflows into the drainage network can lead to air-water interactions that give rise to geysers, causing damages to the water infrastructure and threatening human lives. Over the past two decades, geysers have been attributed to pressure surges caused by rapid inflows (e.g., Guo and Song 1990) and air-water interactions due to instabilities associated with the transition from gravity to pressurized flow in pipe systems (e.g., Hamam and McCorquodale 1982; Li and McCorquodale 1999).

The possibility that air pockets can give rise to significant pressures had also been demonstrated theoretically and experimentally (Martin 1976; Zhou et al. 2002a, b). More recently, systematic investigations have advanced significantly our understanding of the flow caused by entrapped air in pipelines (Wright 2013, 2014; Wright et al. 2011; Vasconcelos and Wright 2006, 2008, 2011). These experiments demonstrated the possibility of different types of air-water interactions as a function of riser and pipe diameters, and rate of increase of inflows into an existing pipeline partially filled with water.

While the previous studies have confirmed the role of entrapped air in causing large pressure transients, the mechanism of geyser formation remains elusive primarily because of the lack of detailed observations and measurements of key variables, e.g., interface velocities and air pocket pressure. In particular, the physical mechanisms under which geysers are formed have not been conclusively reported or elucidated. In a recent mathematical model of a geyser (Vasconcelos et al. 2011), the geyser formation is modeled as a Taylor bubble (Davies and Taylor 1950) rising in the vertical riser. From experiments and physical considerations examined in this paper, the vertical air pocket motion in a geyser differs greatly from that of a Taylor bubble. In addition, the model neglects the dynamic coupling between the air pocket in the horizontal pipeline with that in the vertical riser. Furthermore, most of the previous experiments were performed with an entrapped air pocket in a horizontal pipe with a water-filled riser, without an independent pressure head; such a flow configuration cannot realistically simulate the formation of geysers. The air capsule experiments of

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Lewis (2011) are one of the few cases in which background pressure was preserved through a separate reservoir. More recently, the importance of the ratio of riser to pipeline diameter in geyser formation was revealed by experiments and theory (Wright 2013). However, the theory was based on the assumption that the air pocket is a Taylor bubble and air supply flow arguments without clear physical justifications.

This paper reports a comprehensive experimental investigation aimed to unravel the mechanism of geyser formation in laboratory experiments. An entrapped air pocket is released into a horizontal pipe connected to a vertical shaft filled with water. The subsequent air pocket migration in the horizontal pipe and vertical shaft is measured and the conditions under which a geyser can be formed are studied. Experiments are performed both without and with an independent external pressure head, covering a wide range of riser to pipe diameters. In the following, the experimental program and design are first presented. The observations and findings are first reported for the case of air release into a water-filled riser without an independent pressure head. The measured time variation of free surface and air-water interface in the riser, and the associated air bubble length, velocity, and pressure is presented. The results for air release into a pressurized horizontal pipe are then presented. The conditions for geyser formation from entrapped air pockets are elucidated for the first time via high speed observations and a heuristic theory.

Experiments

Experimental Program

A comprehensive series of experiments has been performed on a physical model of a simplified drainage system, which consists of a 1.8 m high vertical riser and a horizontal pipe with diameter D connected to a constant head tank. A physical model of a simplified drainage system is set up in the hydraulic laboratory as shown in Fig. 1. The model consists of a horizontal circular pipe and a vertical riser, which are made of acrylic material such that the flow phenomenon inside the pipe could be observed and recorded. The pipe is approximately 6 m long with an inner diameter of $D = 0.05$ m. A riser of variable diameter ($D_r = 0.016$ – 0.046 m) is connected to the pipe with a T-junction. Two water tanks are connected to the test section at the upstream and downstream ends, respectively. The upstream tank has a dimension of $0.51 \times 0.63 \times 0.69$ m and is fitted with an overflow weir system inside to provide constant head for the pipe system. The two tanks are connected with a recirculating flow system; the flow in the downstream tank can be transferred back to the constant head tank by a submersible pump. Four PVC quarter-turn ball valves are used to separate the pipe into different sections. The downstream end of the pipe is fitted with a removable cap which allows drainage of the water after an experiment.

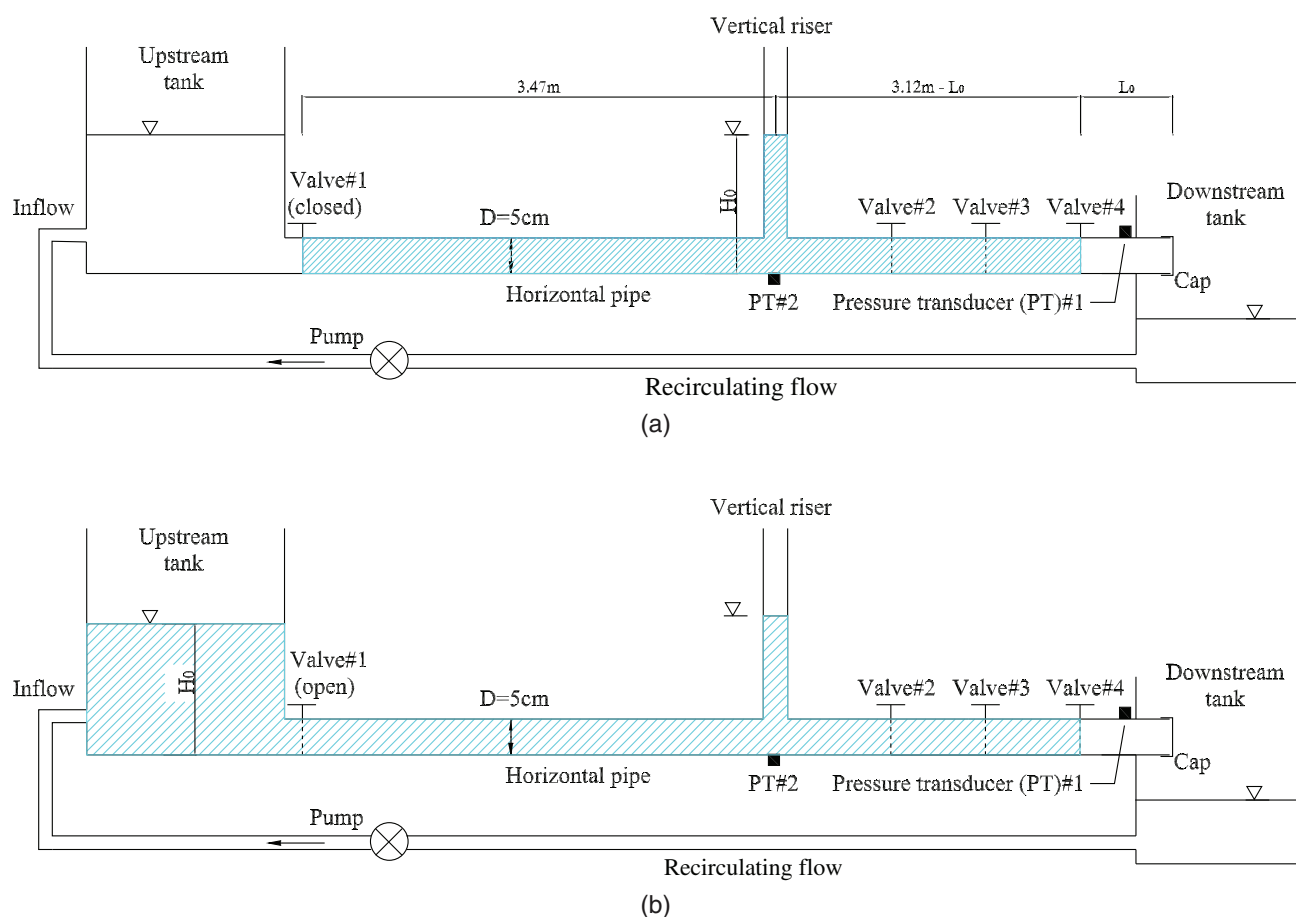


Fig. 1. (Color) Schematic diagram of the experimental apparatus for study of air-water interaction in horizontal pipe with vertical riser, where (a and b) are initial conditions of experiment on release of air pocket from horizontal pipe into vertical riser for Series A and B, respectively: (a) Series A, without an external pressure head; (b) Series B, with an external pressure head

Table 1. Summary of Experimental Results of Series A

Run number	D_r (m)	H_0 (m)	L_0 (m)	T_a (s)	$\frac{U_f}{\sqrt{gD}}$	$\overline{v_{int}}$ (m/s)	$\overline{v_{net}}$ (m/s)	v_{Taylor} (m/s)	$\frac{D_r}{D}$	$\frac{V_{air}}{V_w}$	Geyser
A-1	0.016	0.66	0.61	8.24	0.435	0.22	0.140	0.131	0.320	9.03	No
A-2	0.016	0.77	0.61	8.32	0.431	0.19	0.120	0.131	0.320	7.74	No
A-3	0.016	0.88	0.61	8.24	0.435	0.18	0.110	0.131	0.320	6.77	No
A-4	0.016	0.66	1.2	6.28	0.437	0.28	0.130	0.131	0.320	17.76	No
A-5	0.031	0.66	0.61	8.2	0.437	0.21	0.174	0.182	0.620	2.40	No
A-6	0.031	0.77	0.61	8.24	0.435	0.22	0.185	0.182	0.620	2.06	No
A-7	0.031	0.88	0.61	8.2	0.437	0.2	0.181	0.182	0.620	1.80	No
A-8	0.031	0.66	1.2	6.4	0.428	0.24	0.182	0.182	0.620	4.73	No
A-9	0.041	0.66	0.61	8.24	0.435	0.22	0.197	0.207	0.800	1.37	No
A-10	0.041	0.77	0.61	8.2	0.437	0.22	0.215	0.207	0.800	1.18	No
A-11	0.041	0.88	0.61	8.24	0.435	0.23	0.208	0.207	0.800	1.03	No
A-12	0.041	0.66	1.2	6.4	0.428	0.25	0.200	0.207	0.800	2.70	No
A-12	0.041	0.66	1.2	6.4	0.428	0.25	0.200	0.207	0.800	2.70	No
A-13	0.016	0.88	1.8	4.5	0.419	0.48	0.174	0.131	0.320	19.98	No

Note: T_a = time when air pocket arrives at the riser after valve opening; U_f = average velocity of air pocket front measured in horizontal pipe; V_{air} = initial air volume; V_w = initial water volume in riser; $\overline{v_{int}}$ = average measured velocity of the air pocket front in vertical riser; $\overline{v_{net}}$ = average net velocity of the air pocket front in vertical riser; v_{Taylor} = corresponding Taylor bubble velocity.

Table 2. Summary of Experimental Results of Series B

Run number	D_r (m)	H_0 (m)	L_0 (m)	T_a (s)	$\frac{U_f}{\sqrt{gD}}$	$\overline{v_{fs}}$ (m/s)	$\overline{v_{int}}$ (m/s)	$\overline{v_{net}}$ (m/s)	v_{Taylor} (m/s)	$\frac{D_r}{D}$	$\frac{V_{air}}{V_w}$	Geyser
Video camera												
B-1	0.016	0.66	0.61	8.48	0.423	0.975	1.450	0.475	0.131	0.32	9.03	Yes
B-10	0.016	0.77	0.61	8.68	0.413	0.796	1.189	0.393	0.131	0.32	7.74	Yes
B-11	0.016	0.88	0.61	8.4	0.437	0.567	1.012	0.445	0.131	0.32	6.77	Yes
B-12	0.016	0.66	1.2	6.68	0.410	1.152	1.455	0.304	0.131	0.32	17.76	Yes
B-13	0.016	0.66	1.8	4.68	0.403	1.583	2.018	0.435	0.131	0.32	26.63	Yes
B-14	0.031	0.66	0.61	8.44	0.425	0.306	0.522	0.215	0.190	0.62	2.40	No
B-23	0.031	0.77	0.61	8.36	0.429	0.116	0.323	0.207	0.190	0.62	2.06	No
B-24	0.031	0.88	0.61	8.32	0.431	0.073	0.283	0.209	0.190	0.62	1.80	No
B-25	0.031	0.66	1.2	6.52	0.421	0.579	0.810	0.231	0.190	0.62	4.73	Yes
B-26	0.031	0.66	1.8	4.72	0.399	0.515	0.750	0.235	0.190	0.62	7.09	Yes
B-32	0.041	0.66	0.61	8.32	0.431	0.290	0.529	0.239	0.216	0.82	1.37	No
B-41	0.041	0.77	0.61	8.32	0.431	0.219	0.462	0.242	0.216	0.82	1.18	No
B-42	0.041	0.88	0.61	8.28	0.433	0.120	0.354	0.234	0.216	0.82	1.03	No
B-43	0.041	0.66	1.2	6.6	0.415	0.368	0.614	0.246	0.216	0.82	2.70	No
B-44	0.041	0.66	1.8	4.68	0.403	0.323	0.561	0.239	0.216	0.82	4.06	No
High speed camera												
B-H1	0.016	0.66	0.61	8.07	0.444	0.924	1.231	0.301	0.131	0.32	9.03	Yes
B-H2	0.021	0.66	0.61	7.84	0.457	0.768	1.022	0.261	0.157	0.42	5.24	Yes
B-H3	0.026	0.66	0.61	8.18	0.438	0.657	0.916	0.267	0.174	0.52	3.42	Yes
B-H4	0.031	0.66	0.61	8.14	0.440	0.207	0.418	0.215	0.190	0.62	2.40	No
B-H5	0.036	0.66	0.61	8.1	0.443	0.261	0.481	0.228	0.205	0.72	1.78	No
B-H6	0.041	0.66	0.61	8.1	0.443	0.246	0.476	0.235	0.219	0.82	1.37	No
B-H7	0.046	0.66	0.61	8.22	0.436	0.203	0.441	0.244	0.232	0.92	1.09	No

Note: $\overline{v_{fs}}$ = average measured velocity of the free surface in vertical riser.

In an experiment without external pressure [Series A, Fig. 1(a)], the riser and horizontal pipe are filled with water to an initial head of H_0 ; the downstream section of the pipe has an air pocket of length L_0 and volume V_{air} . By opening or closing a selected ball valve, air pockets of different initial lengths (or volumes) can be formed. Conversely, most experiments are conducted with a pressurized pipe connected to the constant head tank [Series B, Fig. 1(b)] with the initial water depth in the riser at the same level. A total of 22 experiments on geyser formation have been conducted with seven riser diameters of $D_r = 0.016$ – 0.046 m, $H_0 = 0.57$ – 0.88 m, and $L_0 = 0.61$ – 1.8 m. A summary of the experimental parameters is given in Tables 1 and 2.

Experimental Procedure and Measurements

In a typical experiment, a selected ball valve is closed, and the pipe between the valve and the upstream tank is filled with water to a predetermined level in the vertical riser as well as in the tank, measured with a ruler attached to the riser. The water between the valve and the downstream end is then emptied. The downstream end of the pipe is then closed with a plastic cap, leaving an air pocket at atmospheric pressure. The ball valve is then manually opened quickly, and the subsequent motion of the air pocket recorded by video; the manual operation takes approximately 0.2 s. This period is less than the measured pressure oscillation cycle (see later discussion). The experiment

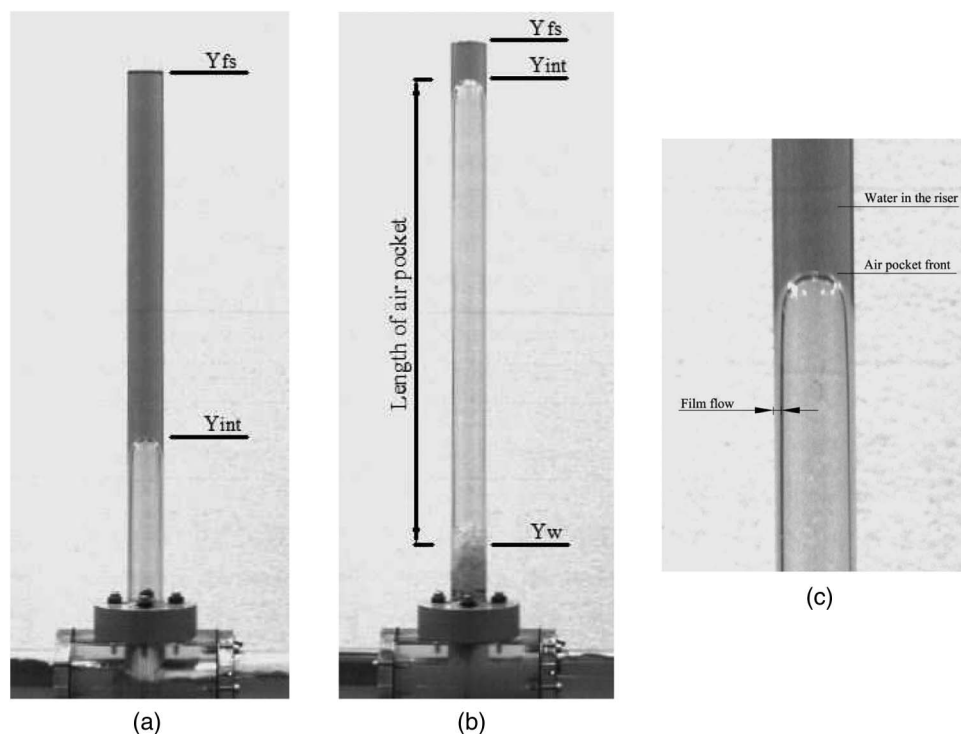


Fig. 2. Definition of air pocket rising in vertical riser: (a) air pocket without tail; (b) air pocket with tail; (c) front of Taylor bubble

duration for each run is approximately 20 s; every experiment is repeated at least once to ensure the repeatability of the experimental results.

During the initial stages of the investigation, two digital HD video camera recorders (Sony, model HDR-SR1E, speed 25 frames per second, resolution 1920×1080 pixels) are used to record the transient process. The horizontal air pocket propagation in the pipe between the closed end and the riser entrance (approximately 1.8 m) is recorded with one camera. A second camera is used to record the air-water interface movement in the vertical riser. Both cameras are positioned at approximately 2 m from the horizontal pipe. In the later stages of the experiment, the detailed air-water flow near the riser-pipe junction and within the vertical riser is also recorded with a high speed camera (DANTEC DYNAMICS SpeedSense 9040, Phantom V9, resolution 1632×1200 pixels) at 100 frames per second (fps). For the measurement of air-water interfaces in the riser, the entire length of the riser needs to be covered within the photo; a Nikon AF NIKKOR 20 mm f/2.8D lens is used to obtain a wider angle of view. The camera speed is set at 50 fps to obtain a balance of field of view, light exposure and measurement accuracy.

After the experiment, the instantaneous images obtained by high speed camera (for video camera cases, the images can be extracted from the video at 0.04 s intervals by *Adobe Premiere CS6*) are imported into *AutoCAD* for postprocessing. In *AutoCAD*, the measurement in all the images is calibrated by the actual length of the riser (1.8 m). For the vertical riser, the distance between the free surface level (Y_{fs}), air pocket front level (Y_{int}), and air pocket tail level (Y_w) to the riser entrance are measured from the image and recorded into spreadsheets (Fig. 2). The sampling interval is approximately 0.2 s for the video camera and 0.06 s for the high speed camera. Thus the time variation of these quantities can be obtained. As the position of the interface is determined visually, the accuracy of the measurement primarily depends on the sharpness of the interface; the estimated accuracy

of interface determination is within 4 and 1 mm for the video and high speed camera images respectively. The propagation speed of the air pocket in the horizontal pipe is found to be quite steady; the average speed is obtained by dividing the length and time of travel between valve and vertical riser. For the analysis of air compression effect (see later discussion), both the air pocket front and reflected wave front in the horizontal pipe need to be observed simultaneously together with the air-water interface in the riser around the time of air entry into the riser. A video camera is used to specially observe the flow at the riser-pipe junction for this purpose; the length of the horizontal pipe and vertical riser sections covered by this video is approximately 1 and 1.8 m, respectively. The average air pocket front speed in the horizontal pipe, free surface, and interface velocities in the riser are given in Tables 1 and 2.

To measure the pressure variation inside the pipe, two piezoelectric pressure transducers (Honeywell model AB/HP 006PSI) are connected to the pipe at different locations. One pressure transducer (PT1) is mounted at the pipe crown near the pipe end to measure the air pressure before the water filled the empty pipe and the other one (PT2) is mounted at the pipe invert at the bottom of the riser to measure the water pressure (Fig. 1). The measurement range is from 0 to 41.4 kPa, with a maximum pressure head of approximately 4.23 m, which is sufficient for these experiments. The data are collected by a data logger (*Scientific Interface 500*) at a frequency of 100 Hz and imported into spreadsheets afterward. The indoor temperature of the laboratory is measured as 23°C, thus the corresponding water density ρ_w is 998 kg/m³. Additional details of the experiments can be found in Cong (2016).

Results and Analysis

The run parameters and results for experiments with air pocket release through water-filled riser (without independent prescribed

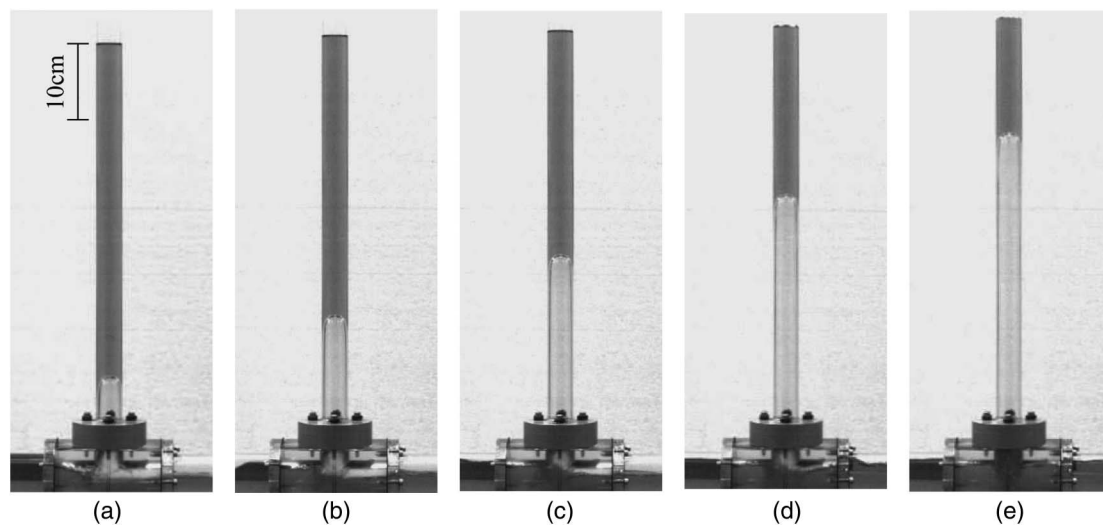


Fig. 3. Observed rising motion of air pocket in a vertical riser (Run A-7): (a) $t = 8.7$ s; (b) $t = 9.1$ s; (c) $t = 9.5$ s; (d) $t = 9.9$ s; (e) $t = 10.3$ s

head) are listed in Table 1. The experimental parameters for runs with an independent pressure head are listed in Table 2.

Air Pocket Release through Water-Filled Riser (Series A)

In these experiments, the horizontal pipe and vertical riser are first filled with water; an air pocket in a section of horizontal pipe is then released through vertical riser; the pressure variation and air-water interaction in vertical riser are measured. Three riser diameters ($D_r = 0.016, 0.031$, and 0.041 m), three initial air lengths ($L_0 = 0.61, 1.2$, and 1.8 m), and three initial water levels ($H_0 = 0.66\text{--}0.88$ m) are varied to investigate their effect on the air-water interaction process. This experiment is similar to the configuration in the study of Vasconcelos (2011) in that the pressure on the pipe is not independent and closely linked with the air-water interaction. A number of flow features are observed:

- The free surface water level in the vertical riser undergoes a strong oscillation, immediately after the opening of the valve. An air slug starts to migrate toward the vertical riser and water flows to fill up the emptied section. A gravity current is developed. The air pocket advances toward the riser entrance at a steady speed; the front has an angle of approximately 60 degrees to the horizontal, with a velocity of $U_f/\sqrt{gD} \approx 0.44$;
- The amplitude of the oscillation is damped quickly before the air pocket reaches the vertical riser;
- As the air pocket rises in the vertical riser, two types of air pocket shape in the vertical riser are observed. At the early stage of rise, a single interface (Y_{int}) can be observed, and the tail of the air pocket is still connected to the residual air in the horizontal pipe [Fig. 2(a)]. When all of the air pocket enters the riser, and the entrapped air pocket volume is smaller than the initial water volume in the riser, a trailing air-water interface (Y_w) is formed [Fig. 2(b)]; and
- The shape of the air pocket is similar to a slug flow or Taylor bubble (Davies and Taylor 1950; Wallis 1969). As the air rises it occupies the center of the riser, while the water flows downward in the form of a thin film along the perimeter wall of the riser [Fig. 2(c)]. Fig. 3 shows the observed motion of an air pocket in the vertical riser; the steady rise of the air-water interface is shown. No strong oscillation of water

or air-water mixture jetting out of riser (geyser event) is observed.

Fig. 4(a) shows a typical result on the measured variation of the free surface and interface level with time (Run A-8). After approximately 6.3 s propagation in the horizontal pipe, the air pocket enters the riser, and starts to rise due to buoyant acceleration. It is

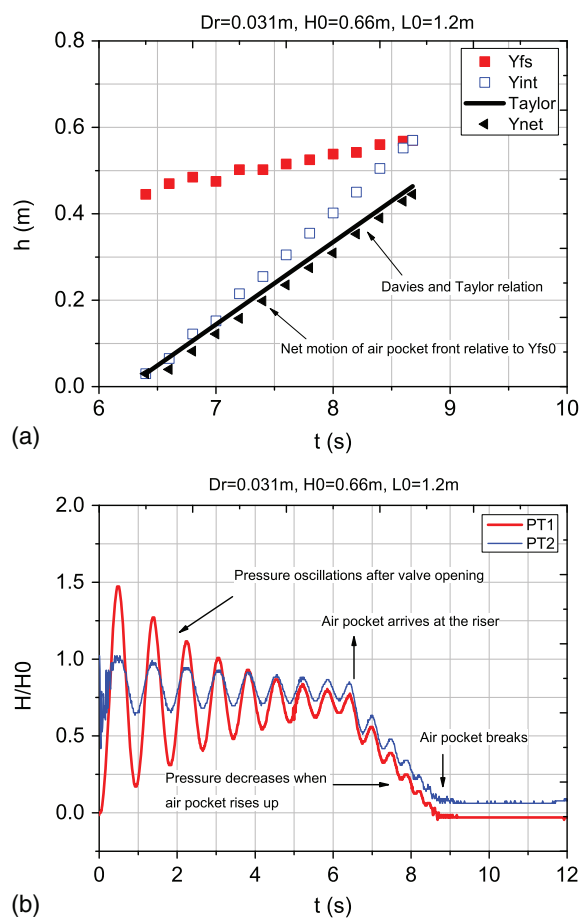


Fig. 4. (Color) Experimental results for Run A-8: (a) temporal variation of air-water interface locations; (b) pressure oscillation pattern

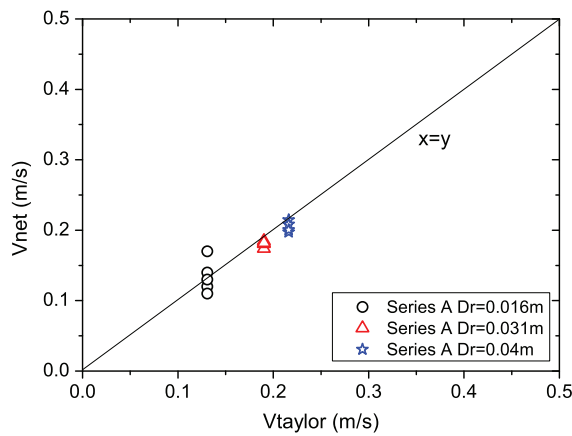


Fig. 5. (Color) Comparison between v_{net} and v_{Taylor} : Series A

observed that Y_{int} migrates much more rapidly than Y_{fs} during the rising process. The free surface undergoes small oscillation and stays nearly constant during the ascent of the air pocket. The water between Y_{fs} and Y_{int} flows downward in the film around the air pocket. The air pocket front Y_{int} eventually catches up with the free surface Y_{fs} as the air pocket breaks. Based on the measured trajectories, the average velocity of Y_{fs} and Y_{int} can

be obtained. As the free surface moves, there is an upward water flow in the riser during the air pocket rise; the measured velocity of the air pocket is hence influenced by both buoyancy and the water flow in the riser. To account for the influence of the ambient water flow on the air pocket motion, the net motion of the air pocket is defined as $Y_{\text{net}} = Y_{\text{fs}0} - (Y_{\text{fs}} - Y_{\text{int}})$ in which $Y_{\text{fs}0}$ is the free surface level at the time when the air pocket is about to rise in the riser.

For the present experiments, surface tension effects can be assessed by considering the Eötvös number defined as $N_{\text{Eo}} = \rho_w g D_r^2 / \sigma$, where D_r = riser diameter; and σ = surface tension of water = 0.072 N/m. For the riser diameters used in this study $D_r = 0.016\text{--}0.045$ m, $N_{\text{Eo}} = 35\text{--}276$, which is at least one order of magnitude greater than the critical threshold value of $N_{\text{Eo}} = 3.37$ (Wallis 1969), below which surface tension effects will be important in the air pocket motion. Similarly it can be shown that the equilibrium air pocket rising velocity can be given by a semi-empirical relation $v_{\text{Taylor}} = 0.345 \sqrt{g D_r}$ (Wallis 1969; Cong 2016). Fig. 5 shows the comparison between $v_{\text{net}} = dY_{\text{net}}/dt$ and V_{Taylor} for all the experiments. It can be concluded that the net speed of the air pocket is in close agreement with the theoretical Taylor bubble velocity. From the images the thin film thickness can also be estimated to be 0.6–1.2 mm, which agrees well with the two-dimensional (2D) film flow theory of Batchelor (1967), with $\delta/a = 0.9(\nu^2/a^3g)^{1/6}$ (where δ = film thickness; and $a = D_r/2$ is the riser radius).

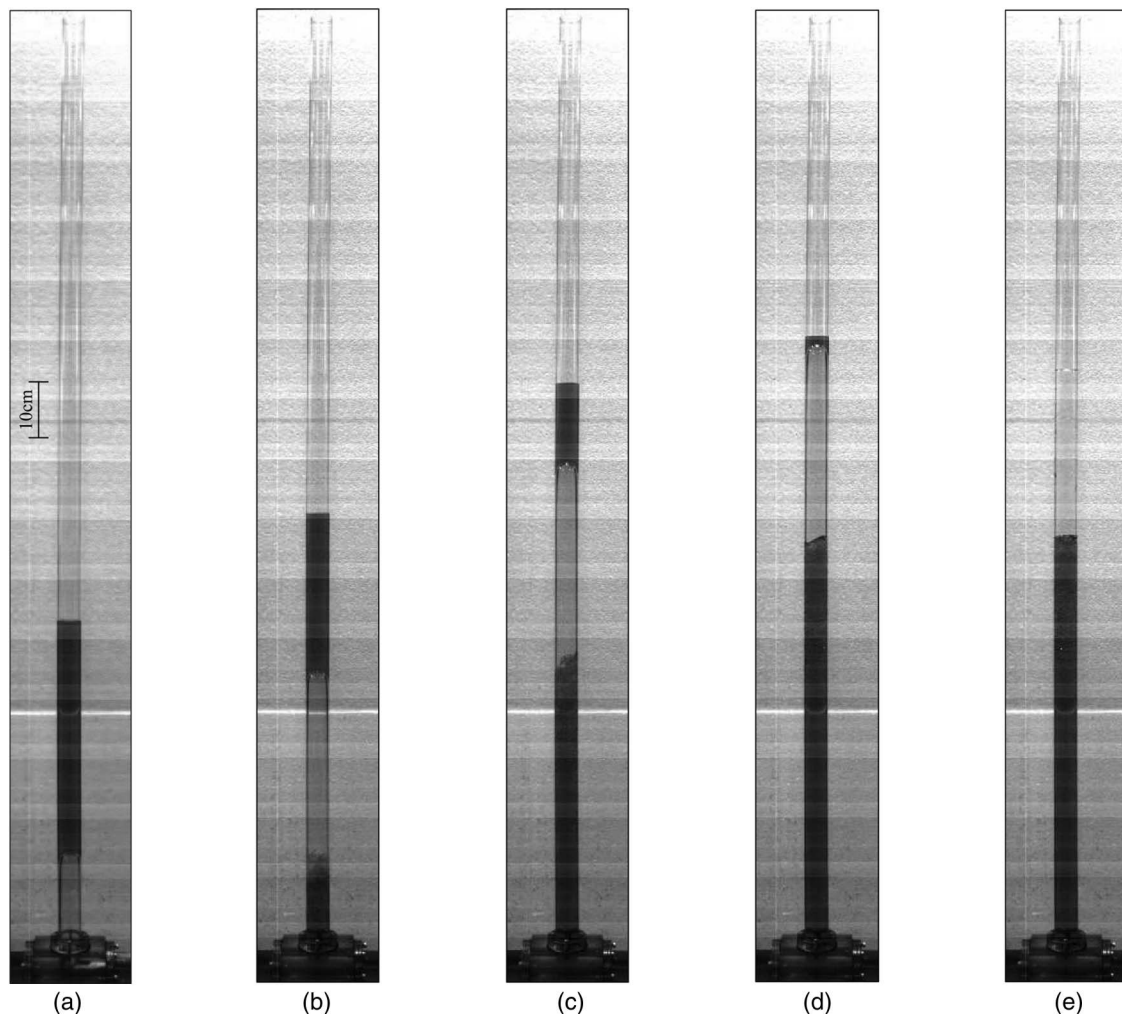


Fig. 6. Instantaneous frames of air pocket rising in a vertical riser (Run B-H6): (a) $t = 8.7$ s; (b) $t = 9.3$ s; (c) $t = 9.9$ s; (d) $t = 10.5$ s; (e) $t = 10.9$ s

Fig. 4(b) shows the measured normalized pressure variation H/H_0 . Basically the pressure measured by PT1 has the same period but larger amplitude than that measured by PT2. Significant pressure oscillations induced by the initial pressure difference are measured after valve opening. The amplitude of the oscillation decreases before the air arrives at the riser. As the air pocket rises the pressure decreases as the water between Y_{fs} and Y_{int} is running away through the film between the wall and the air pocket. After the air pocket breaks the pressure is the lowest and stays constant. In general, the amplitude of the pressure oscillations decreases with riser diameter due to friction damping.

The temporal variation of the air-water interface and pressure is basically similar for all the experiments; additional details can be found in Cong (2016).

Air Pocket Release through Vertical Riser in Pressurized Horizontal Pipe (Series B)

For these experiments, unlike the Series A experiments, an independent pressure head is imposed on the horizontal pipe-vertical riser system. The summary of experimental parameters and key results are presented in Table 2. Seven riser diameters ($D_r = 0.016, 0.021, 0.026, 0.031, 0.036, 0.041$, and 0.046 m), three initial air lengths ($L_0 = 0.61, 1.2$, and 1.8 m), and three initial water levels

($H_0 = 0.66\text{--}0.88$ m) are varied to investigate their effect on the air-water interaction process.

Observation of Geysers

In some experiments, geysers with air-water interaction are observed. In this study a geyser is defined as the ejection or splash of air-water mixture through the top of the vertical riser. It is a result of rapid vertical acceleration of an air pocket, which rapidly lifts the water column above it. The height of the riser used in this study is 1.8 m (measured from the soffit of the horizontal pipe), which is approximately two times the maximum initial pressure head $H_0 = 0.88$ m. While in other experiments, no geyser is observed, and the air-water interaction ends with the breaking of air pockets through the free surface.

Fig. 6 shows the observed rise motion of the air pocket in an experiment without geysers (Run B-H6). The air pocket arrives at the riser at approximately $t = 8.6$ s and rises in the riser steadily, very similar to that of a Taylor bubble; the air pocket breaks within the riser when it catches up with the free surface. The measured variation of the air-water interfaces in the vertical riser is shown in Fig. 7 along with the vertical air pocket speed v_{int} , the length of water column above the air pocket L_w , as well as the measured pressure head of the air pocket H_a (see later discussion). It is shown that the measured air pocket velocity relative to the free

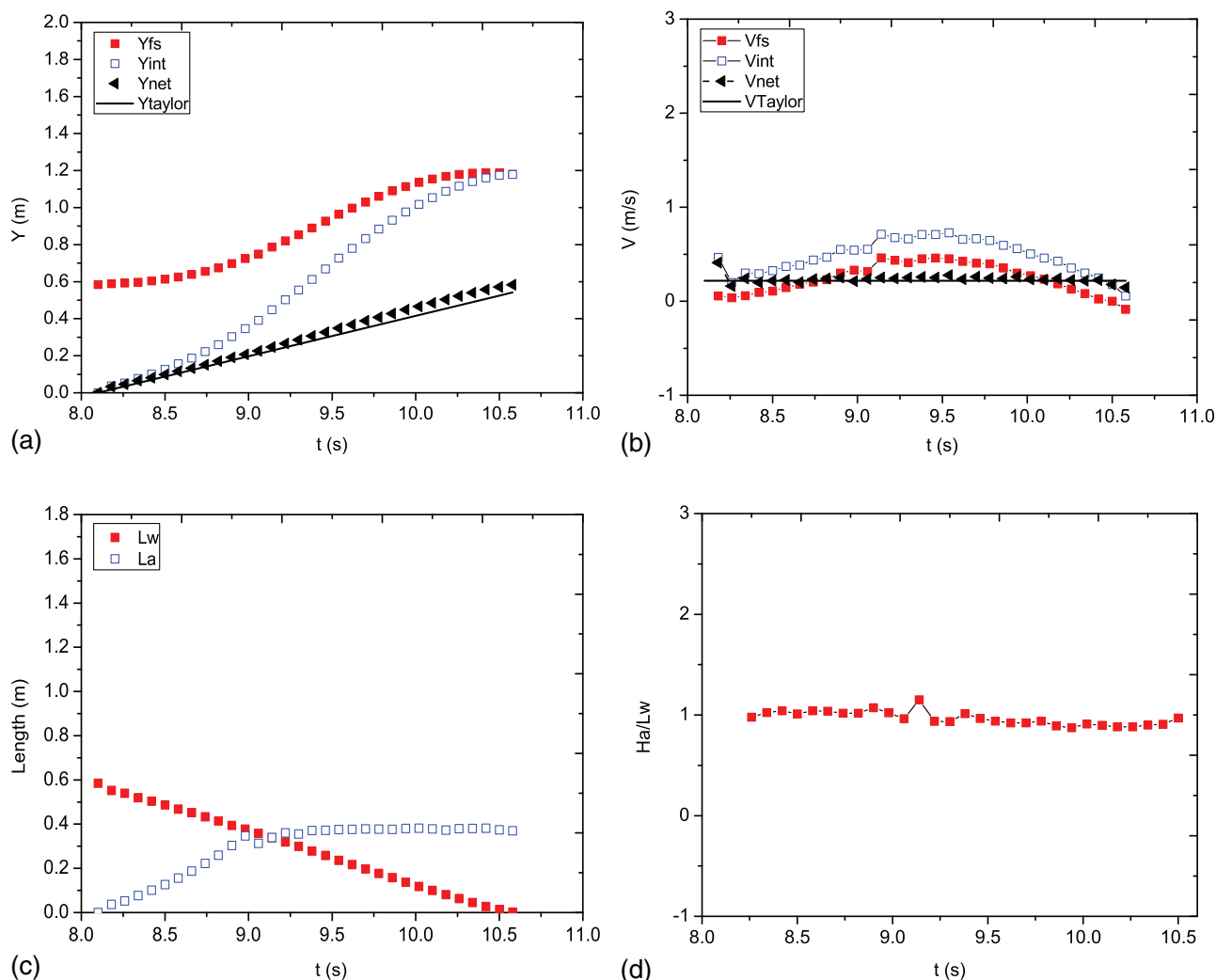


Fig. 7. (Color) Experimental results of Run B-H6: $D_r = 0.041$ m, $H_0 = 0.66$ m, $L_0 = 0.61$ m ($D_r/D = 0.82$, $V_{air}/V_w = 1.37$)

surface v_{net} is close to that of the Taylor bubble velocity v_{Taylor} [Figs. 7(a and b)]. Note that the length of the water column in the riser decreases steadily with time, the air pocket pressure is given by the hydrostatic head of the water column, and the air pocket length L_a approaches an asymptotic value.

Conversely, Figs. 8 and 9 show the observed rise motion of the air pocket and the measured air-water interface variation with time, respectively, for an experiment with geysers (Run B-H1). The air pocket arrives at the riser bottom approximately 8.5 s after the valve opening. Only part of the air pocket enters the riser; during the ascent the pocket nose Y_{int} undergoes a very fast acceleration. The free surface Y_{fs} is pushed by the air to rise rapidly to the riser top, which results in water being ejected out from the riser. Both

the Y_{fs} and Y_{int} increase rapidly with time. Note that the air bubble velocity is far greater than the Taylor bubble velocity during the geyser event [Figs. 9(a and b)].

For the pressure measurement during the process (Fig. 10), a sinusoidal pressure oscillation is observed after the valve opening, which is attributable to the compression and depression of air in confined pipe. When the air pocket arrives at the riser, the pressure starts to decrease as the air is replacing the water in the riser. In contrast to the experiments without an external pressure head, in this experiment, after a short time of decreasing (approximately 1 s), the pressure head dramatically increases to approximately $1.9 \times H_0$ quickly and causes a geyser [Fig. 10(a)]. For the experiment without a geyser a smaller pressure oscillation with larger period

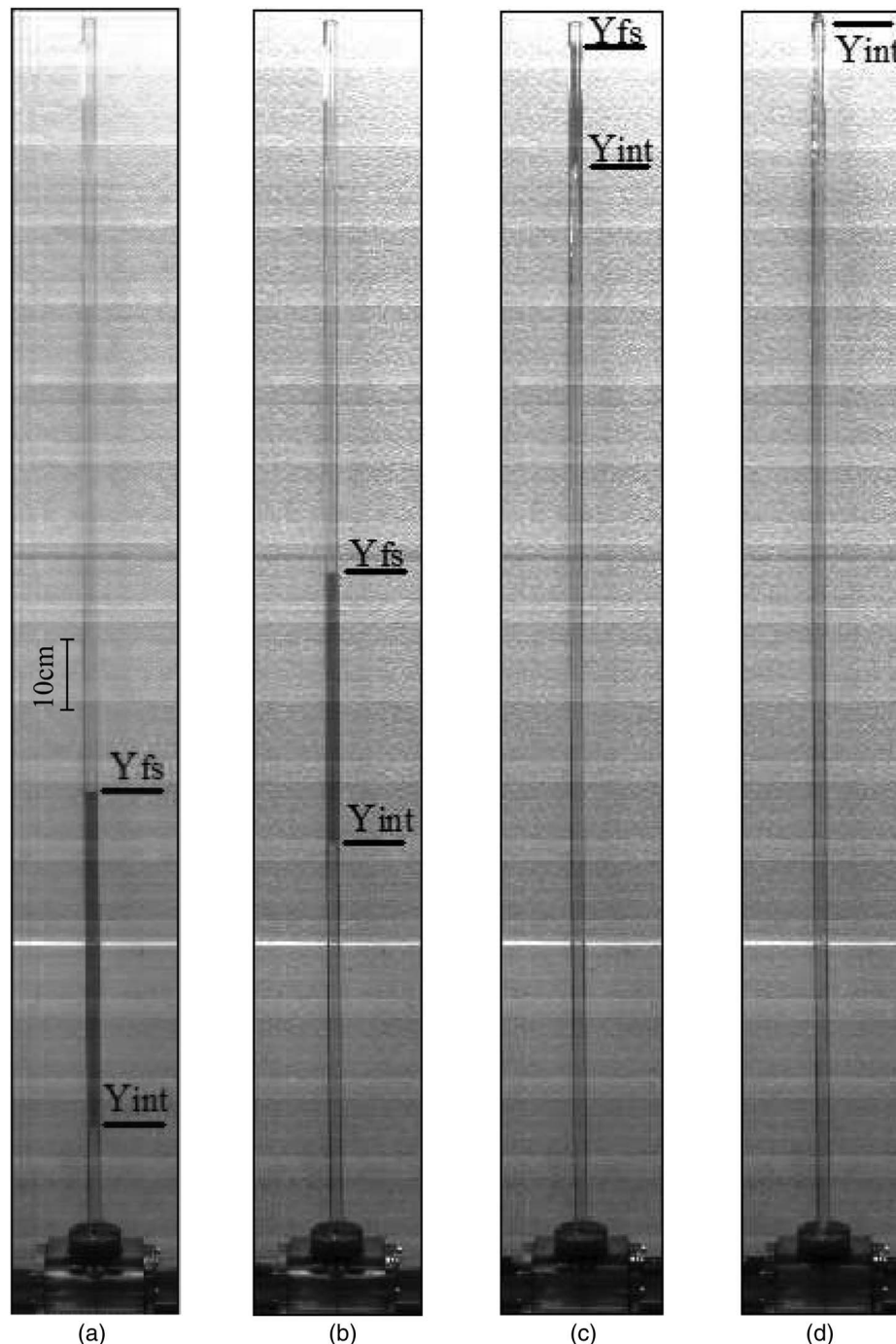


Fig. 8. Instantaneous frames of air pocket rising in a vertical riser (Run B-H1): (a) $t = 8.67$ s; (b) $t = 9.07$ s; (c) $t = 9.47$ s; (d) $t = 9.55$ s

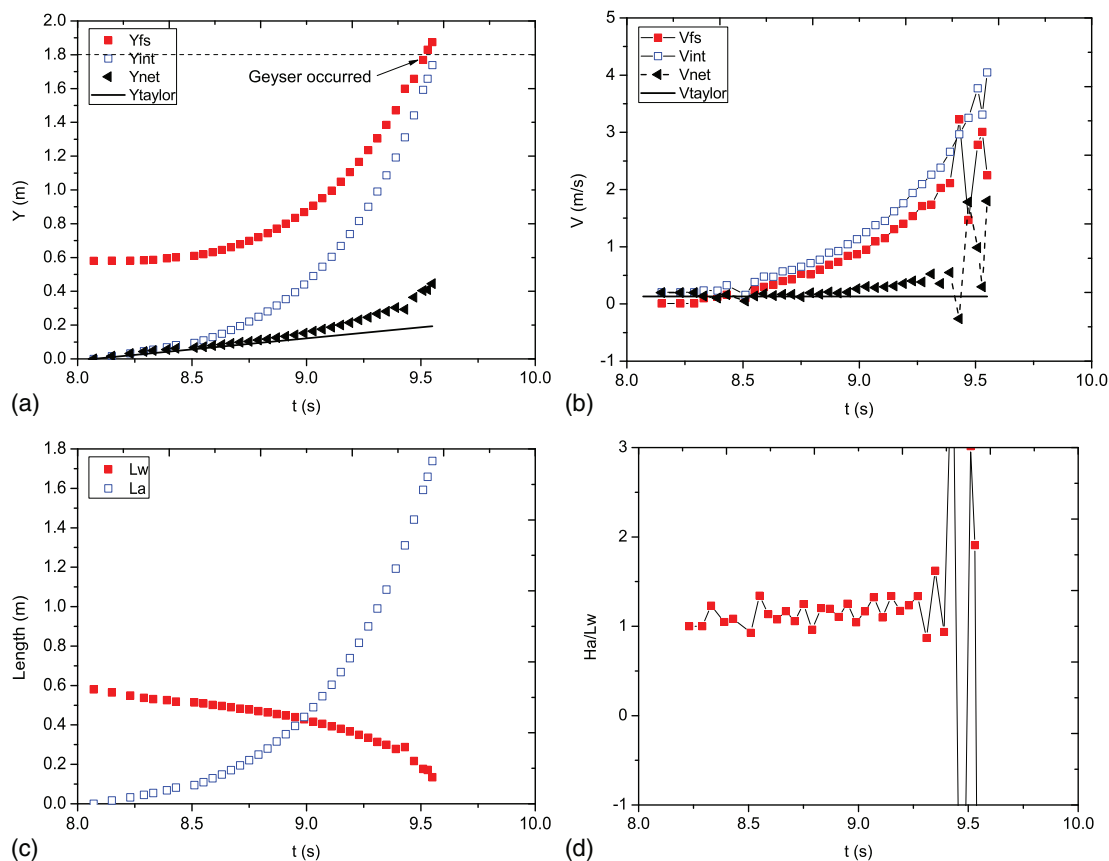


Fig. 9. (Color) Experimental results of Run B-H1: $D_r = 0.016$ m, $H_0 = 0.66$ m, $L_0 = 0.61$ m ($D_r/D = 0.32$, $V_{air}/V_w = 9.03$)

is observed during the rising of air pocket, and the peak pressure head is approximately $1.4 \times H_0$ [Fig. 10(b)]. The only difference between these two experiments (Run B-H1 and B-H6) is the riser diameter, $D_r = 0.016$ and 0.041 m. It can be concluded that geyser is more likely to occur in a smaller riser. The pattern of pressure drop and subsequent surge during a geyser occurrence in these experiments is remarkably similar to that observed in real systems (e.g., the pressure measurements during the geyser episode observed in the Minneapolis, Minnesota stormwater system as reported in Wright 2014). Other representative experimental results can be found in the Supplemental Data (Figs. S1–S5) and in Cong (2016).

The effect of the air volume on the geyser occurrence is studied by the comparison between Runs B-25 and B-14, in which a larger riser diameter is used in these two cases ($D_r = 0.031$ m). With a smaller air pocket ($L_0 = 0.61$ m), no geyser is observed [Fig. 11(a)]. Conversely, with larger air volume in the pipe ($L_0 = 1.2$ m), more air enters the riser and is more likely to generate a geyser [Fig. 11(b)]. The collective experimental results show that geysers are more likely to occur for small risers ($D_r/D \leq 0.62$) and large air volumes $\{V_{air}/[(\pi D_r^2/4)H_0] \geq 3.42\}$ (Table 2).

Estimation of Air Pocket Pressure

Based on observations, the authors propose a theory on the criteria of geyser formation considering the compression of air pocket. For the air-water flow in the vertical riser, it can be simplified as a water column driven by the air pressure above the geyser, the momentum equation of the water column above the air pocket is

$$\frac{d}{dt}(\rho_w L_w A_r v) = p_a A_r - \rho_w L_w A_r g - \frac{f}{8} \rho_w v^2 \pi D_r L_w \quad (1)$$

where ρ_w = density of water; g = gravitational acceleration; L_w = length of water column above the air pocket; A_r = cross-sectional area of riser; p_a = pressure of the air pocket relative to the atmospheric pressure; and f = Darcy-Weisbach wall friction factor ≈ 0.01 (for perspex pipe). The pressure head (in m of water) within the air pocket can therefore be determined by

$$H_a = \frac{p_a}{\rho_w g} = \frac{L_w \left(\frac{dv}{dt} + g + \frac{f}{2D_r} v|v| \right) + v \frac{dL_w}{dt}}{g} \quad (2)$$

from the measured L_w and v .

Mechanism for Geyser Occurrence

Based on observations, the process of a geyser formation can be summarized as in Fig. 12. As a basis for the interpretation of the observations, the measured position and velocity of air pocket front and reflected water wave front (see later discussion) for a typical case with geyser (Run B-1) are shown in Figs. 13(a and b). The air volume in the horizontal pipe and vertical riser [Fig. 13(c)] is calculated by $k_a(X_f - X_r)\pi(D^2/4)$ and $Y_{int}\pi(D_r^2/4)$ (assuming negligible volume for thin film), respectively, where X_f = air pocket front position; X_r = reflected water wave front position; and k_a = the ratio between the cross-sectional area of air pocket and the horizontal pipe. According to current experimental observation, $k_a \approx 0.41$. The observation of a typical geyser process is described as follows:

- At $t \approx 0$ s, after the valve is opened, a density current is formed and the air pocket propagates toward the upstream with an average velocity of $U_f \approx 0.42\sqrt{gD}$, the water wave is reflected when it reaches the pipe end [Fig. 12(a)];

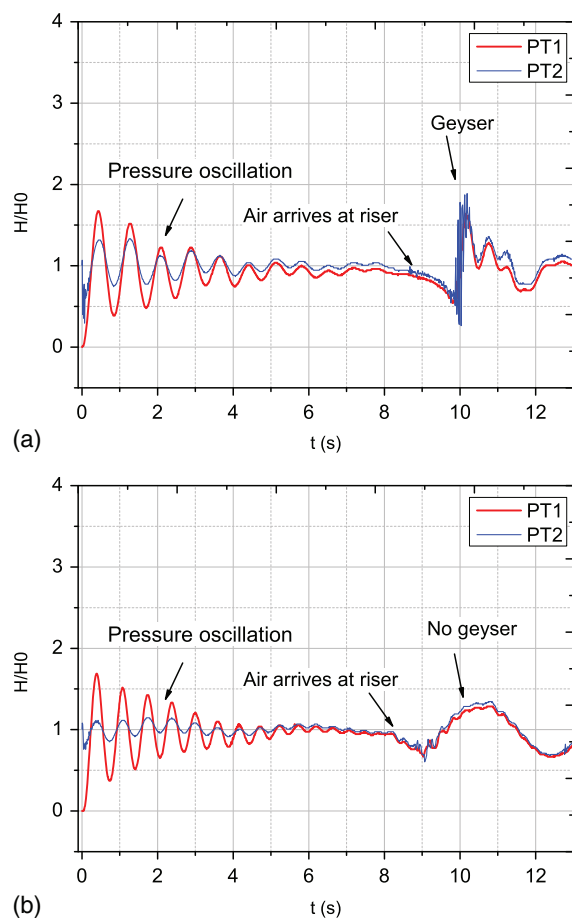


Fig. 10. (Color) Normalized pressure head oscillation: (a) $D_r = 0.016$ m (Run B-1); (b) $D_r = 0.041$ m (Run B-32)

- At $t \approx 8.48$ s, the air pocket front arrives at the riser. As the riser diameter is relatively small, only part of the air pocket makes its way into the riser. The rest of the air pocket continues to advance toward the reservoir. The air pocket that enters the riser initially rises vertically like a Taylor bubble [Figs. 12(b) and 13(a)];
- At $t \approx 9.40$ s, the front velocity of air pocket starts to decrease, the reflected wave front keeps propagating with a fluctuating speed, at this moment $U_r \approx 1.2\sqrt{gD}$. At $t \approx 10.0$ s, the air pocket front nearly stops, and the reflected wave speed $U_r/\sqrt{gD} \approx 0.14$ [Figs. 12(c), 13(a and b)]; and
- After $t \approx 10.0$ s, the air pocket accelerate quickly with air-water mixture jetting out of the riser top [Fig. 12(d)].

After the air pocket enters the riser, the water above the air in the riser falls around the Taylor bubble as a thin film flow and pressure above the air pocket decreases. Thus the pressure difference between the constant upstream pressure and the pressure above the air pocket will prevent the propagation of air front in the horizontal pipe. Meanwhile, the reflected water wave front of the density current also pushes the air from pipe end to the riser. The compression of the air pocket by these two interfaces will pressurize the air pocket in the horizontal pipe and as a result push the air and water out of the riser quickly, with a much larger speed than that of a Taylor bubble (Fig. 12). As shown in Fig. 13(c), from $t = 9.2$ to 9.8 s, the total air volume drop from 0.74 to 0.59 L. According to the ideal gas law $PV = \text{constant}$, a 20% decrease in air volume may give rise to approximately 2.7 m increase in air pressure head (initial air pressure head is 11.06 m before

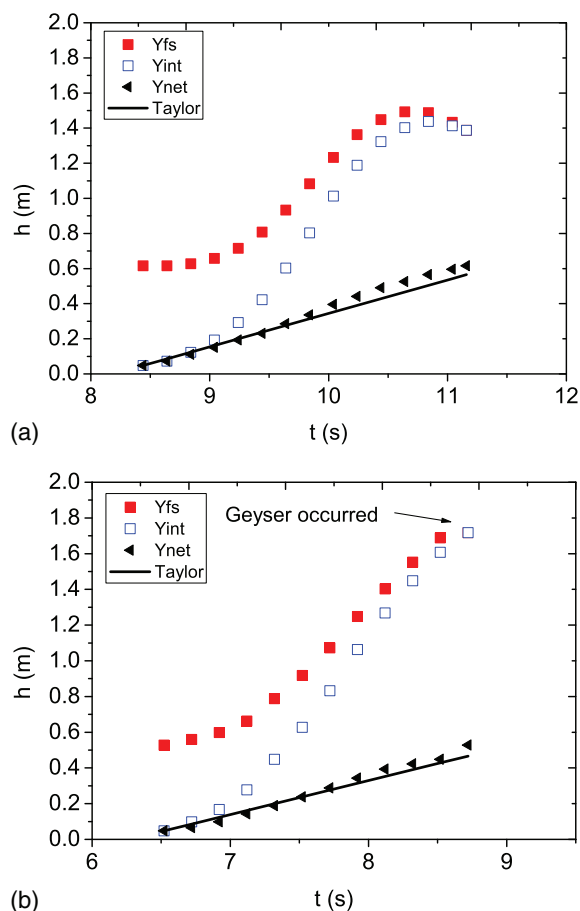


Fig. 11. (Color) Comparison between experiments with different initial air volume: (a) Run B-14: $D_r = 0.036$ m, $H_0 = 0.66$ m, $L_0 = 0.61$ m ($D_r/D = 0.62$, $V_{\text{air}}/V_w = 2.40$); (b) Run B-25: $D_r = 0.036$ m, $H_0 = 0.66$ m, $L_0 = 1.2$ m ($D_r/D = 0.62$, $V_{\text{air}}/V_w = 4.73$)

compression), this pressure is much larger than the water pressure in the riser, and can basically explain the occurrence of geyser phenomenon.

Based on previous analysis, it can be concluded that the geyser formation depends on whether the air pocket will be compressed or not. In principle, air pocket will be compressed if the rate of decrease of air volume in horizontal pipe is larger than the increase of air volume in riser

$$0.41 \times \frac{\pi D^2}{4} k_2 \sqrt{gD} > \frac{\pi D_r^2}{4} \times 0.345 \sqrt{gD_r} \quad (3)$$

where $k_2 = (U_r/\sqrt{gD})$. The previous equation can be simplified as $D_r/D < (1.19k_2)^{0.4}$. When the reflected wave front approaches the riser, its motion is complicated and the velocity fluctuates [Fig. 13(a)]. For $k_2 \approx 0.2$ around the time of geyser occurrence, $D_r/D < 0.56$, which is basically consistent with the experimental results ($D_r/D < 0.62$). Fig. 14 shows that geyser generally occurs when $D_r/D < 0.62$ with v_{net} greater than v_{Taylor} .

Conclusion

Geysers are explosive eruptions of air-water mixture from manholes in drainage systems. Using high speed video and photography, the mechanism for the formation of geysers is elucidated experimentally for the first time in a physical analog

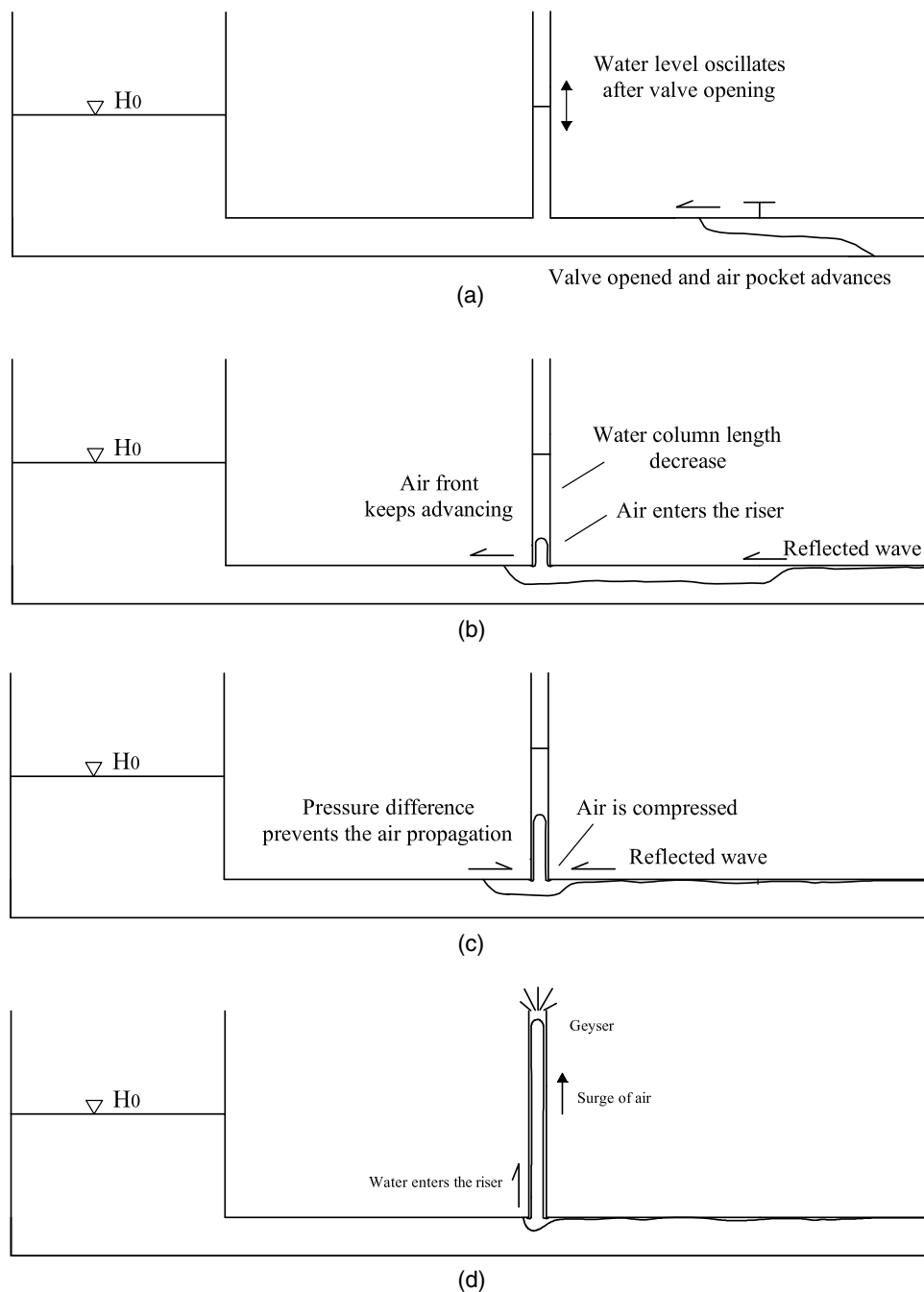


Fig. 12. Observation of a geyser formation process: (a) $0 \text{ s} < t < 8.48 \text{ s}$; (b) $8.48 \text{ s} < t < 9.40 \text{ s}$; (c) $9.40 \text{ s} < t < 10.0 \text{ s}$; (d) $t > 10.0 \text{ s}$

of a highly simplified drainage system. In each experiment a relatively large air pocket (compared with pipe size) is released from a horizontal pipe into a vertical shaft. Without an external pressure head, the air pocket migration in a water-filled vertical riser is found to be similar to a slug flow. The air pocket velocity is close to the Taylor bubble velocity, and no geyser is observed. For a pressurized horizontal pipe, two types of flow are observed. For large riser diameter and small air volume, the air pocket in the vertical riser rises like a Taylor bubble subject to hydrostatic pressure. The air breaks within the riser and no geyser is observed. For small riser diameter and large air volume, the horizontal air pocket shoots past the riser junction and only partially enters the shaft. The subsequent air pocket motion leads to compression of the air pocket and high pressures that

result in rapid acceleration of air and water, and jetting out of air-water mixture from the riser top. A heuristic theory shows that the ratio of riser to pipe diameter D_r/D is a key parameter in determining the occurrence of geysers, with a theoretical threshold of $D_r/D < 0.56$. This is broadly supported by the experiments that show a limiting criterion of $D_r/D \leq 0.62$ and $V_{\text{air}}/[(\pi D_r^2/4)H_0] \geq 3.42$.

This study differs from previous works as follows:

- In the air capsule experiments of Lewis (2011) only crude video-camera measurements of air-pocket and free surface levels are presented, without any detailed analysis. The authors used high speed imaging to accurately observe the air-water interactions over a much wider range of experimental parameters, and pressure transients were also simultaneously measured;

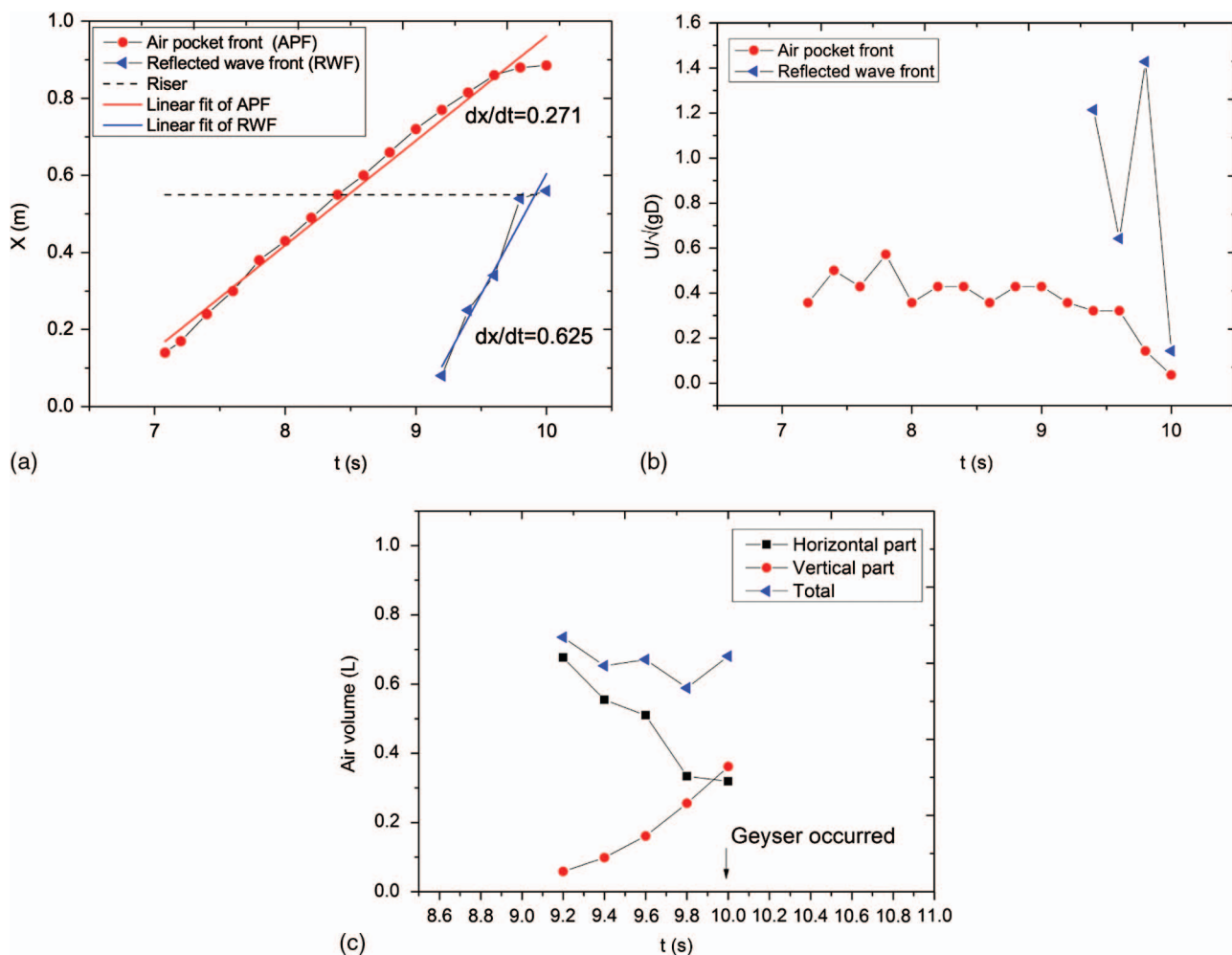


Fig. 13. (Color) Time variation of horizontal air pocket characteristics (Run B-1): (a) location of air front and reflected wave front in horizontal pipe; (b) normalized air front speed and reflected wave front speed in horizontal pipe; (c) temporal variation of air volume

- An alternative definition of geysering (air pocket velocity greater than Taylor bubble velocity) is proposed for the first time. This has helped resolve previous theoretical and experimental modeling discrepancies; and
- The experiments of Vasconcelos and Wright (2011) were performed without an independent pressure head (Series A). It is

shown that the buoyancy effect of the air pocket alone cannot account for the cause of geysers. The present experimental findings are also supported by a detailed three-dimensional (3D) computational fluid dynamics (CFD) study; this will be separately reported.

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Notation

The following symbols are used in this paper:

- $A_r = \pi D_r^2/4$ cross-sectional area of riser;
- D horizontal pipe diameter;
- D_r vertical riser diameter;
- f Darcy-Weisbach wall friction factor;
- $g = 9.81 \text{ m/s}^2$, gravitational acceleration;

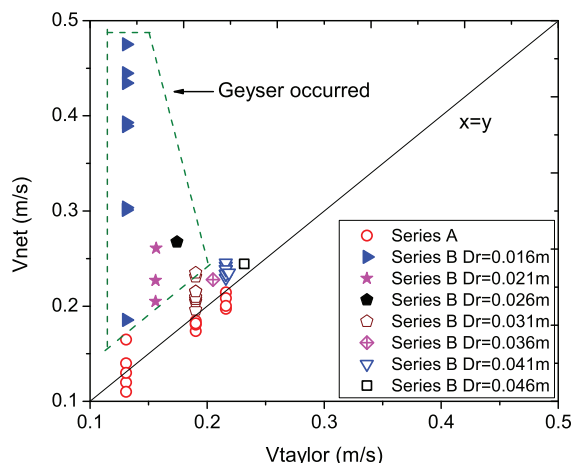


Fig. 14. (Color) Comparison between ν_{net} and ν_{Taylor} : Series A and B

H = measured pressure head at transducers;
 H_0 = initial water levels in the riser (Series A)/upstream head (Series B) from invert of tunnel;
 $H_a = p_a / \rho_w g$ pressure head in air pocket in riser;
 k_a = the ratio between the cross-sectional area of air pocket and the horizontal pipe;
 $k_2 = U_r / \sqrt{gD}$;
 L_a = length of air pocket in the riser;
 L_0 = initial air length in the horizontal pipe;
 $L_w = Y_{fs} - Y_{int}$ length of water column above the air pocket;
 Ne_o = Eotvos number;
 p_a = pressure in air pocket in riser relative to the atmospheric pressure;
 T_a = time when air pocket arrives at the riser after valve opening;
 t = time from the start of experiment;
 U_f = propagation velocity of air front in horizontal pipe;
 U_r = reflected wave speed;
 $V_{air} = \pi D^2 L_0 / 4$ initial air volume;
 $V_w = \pi D_r^2 H_0 / 4$ initial volume of water in the riser;
 $v_{net} = dY_{net} / dt$ net rising velocity of air pocket relative to the free surface;
 v_{Taylor} = velocity of Taylor bubble
 X_f = frontal position of air pocket in horizontal pipe;
 X_r = frontal position of reflected wave in horizontal pipe;
 Y_{fs} = free surface level in the riser;
 Y_{fs0} = free surface level at T_a ;
 Y_{int} = frontal level of air pocket in the riser;
 Y_w = level of air pocket tail in the riser;
 $Y_{net} = Y_{fs0} - (Y_{fs} - Y_{int})$;
 Y_{Taylor} = frontal level of a Taylor bubble;
 δ = film thickness;
 ρ_w = water density;
 $\sigma = 0.072$ N/m surface tension of water; and
 ν = kinematic viscosity of water.

Supplemental Data

The measured time variation of the free surface and air-water interface, and water column and air pocket length in the vertical riser for the additional experiments with $D_r = 0.021, 0.026, 0.031, 0.036$, and 0.046 m, as seen in Figs. S1–S5, are available online in the ASCE Library (www.ascelibrary.org).

References

- Adobe Premiere CS6 [Computer software]. Adobe Systems, Inc., San Jose, CA.
- AutoCAD [Computer software]. Autodesk, Inc., San Rafael, CA.
- Batchelor, G. K. (1967). *Introduction to fluid dynamics*, Cambridge University Press, Cambridge, U.K.
- Cong, J. (2016). *Experimental modeling of air-water interaction in horizontal pipe with vertical riser*, M.Phil. thesis, Hong Kong Univ. of Science and Technology, Hong Kong.
- Davies, R. M., and Taylor, G. I. (1950). "The mechanics of large bubbles rising through extended liquids and through liquids in tubes." *Proc., R. Soc. London, Ser. A*, 200(1062), 375–390.
- Guo, Q., and Song, C. C. S. (1990). "Surging in urban storm drainage systems." *J. Hydraul. Eng.*, 10.1061/(ASCE)0733-9429(1990)116:12(1523), 1523–1537.
- Hamam, M. A., and McCorquodale, A. (1982). "Transient conditions in the transition from gravity to surcharged sewer flow." *Can. J. Civ. Eng.*, 9(2), 189–196.
- Konrad, C. P. (2003). "Effects of urban development on floods, U.S. Geological Survey fact sheet FS-076-03." (<https://pubs.usgs.gov/fs/fs07603/pdf/fs07603.pdf>) (Feb. 1, 2017).
- Lenderink, G., Mok, H. Y., Lee, T. C., and van Oldenborgh, G. J. (2011). "Scaling and trends of hourly precipitation extremes in two different climate zones—Hong Kong and the Netherlands." *Hydrol. Earth Syst. Sci.*, 15(9), 3033–3041.
- Lewis, J. W., (2011). "A physical investigation of air/water interactions leading to geyser events in rapid filling pipelines." Ph.D. thesis, Univ. of Michigan, Ann Arbor, MI.
- Li, J., and McCorquodale, A. (1999). "Modeling mixed flow in storm sewers." *J. Hydraul. Eng.*, 10.1061/(ASCE)0733-9429(1999)125:11(1170), 1170–1180.
- Martin, C. S. (1976). "Entrapped air in pipelines." *Proc., 2nd Int. Conf. on Pressure Surges, British Hydromechanics Research Association*, Vol. 2, City Univ., London, 15–27.
- Science Workshop 500 Interface [Computer software]. PASCO Scientific, Roseville, CA.
- Vasconcelos, J. G., and Wright, S. J. (2006). "Mechanisms for air pocket entrapment in stormwater storage tunnels." *Proc., World Water and Environmental Resources Congress*, Omaha, NE, 40856–14275.
- Vasconcelos, J. G., and Wright, S. J. (2008). "Rapid flow startup in filled horizontal pipelines." *J. Hydraul. Eng.*, 10.1061/(ASCE)0733-9429(2008)134:7(984), 984–992.
- Vasconcelos, J. G., and Wright, S. J. (2011). "Geysering generated by large air pockets released through water-filled ventilation shafts." *J. Hydraul. Eng.*, 10.1061/(ASCE)HY.1943-7900.0000332, 543–555.
- Wallis, G. B. (1969). *One-dimensional two-phase flow*, McGraw Hill, New York.
- Wright, S. J., (2013). "Influence of air pocket volume on manhole surge." *J. Water Manage. Model*, R246-09.
- Wright, S. J. (2014). "Modeling rapid filling processes in stormwater tunnel systems." *Proc., DSD 25th Anniversary Int. Conf.*, Hong Kong Drainage Services Dept., Hong Kong.
- Wright, S. J., Lewis, J. W., and Vasconcelos, J. G. (2011). "Geysering in rapidly filling storm-water tunnels." *J. Hydraul. Eng.*, 10.1061/(ASCE)HY.1943-7900.0000245, 112–115.
- Zhou, F., Hicks, F., and Steffler, P. M. (2002a). "Observations of air-water interaction in a rapidly filling horizontal pipe." *J. Hydraul. Eng.*, 10.1061/(ASCE)0733-9429(2002)128:6(635), 635–639.
- Zhou, F., Hicks, F., and Steffler, P. M. (2002b). "Transient flow in a rapidly filling horizontal pipe containing trapped air." *J. Hydraul. Eng.*, 10.1061/(ASCE)0733-9429(2002)128:6(625), 625–634.